

Congruence and primes



Fix an integer $m > 0$, for any $a \in \mathbb{Z}$.

We may write $a = qm + r$

for $0 \leq r < m$ in a unique way

This defines a function

$$(-) \bmod m : \mathbb{Z} \rightarrow \{0, 1, 2, \dots, m-1\}$$

sending a to the remainder after

quotienting m . We note the preimages

partitions $\mathbb{Z}$ and thus induces an

equivalence relation on $\mathbb{Z}$. This leads

to the study of modular classes.

Def 1: Let $m > 0$. For integers a, b .

we say a is congruent to b modulo m

if they have the same remainder

after quotienting by m . which

we denote by $a \equiv b \pmod{m}$.

lem 2: Let $m > 0$. For $a, b \in \mathbb{Z}$,

$$a \equiv b \pmod{m} \text{ iff } m \mid a - b.$$

proof. ($\Rightarrow$) Suppose $a \equiv b \pmod{m}$.

Then we may write $a = q_1 m + r$

and $b = q_2 m + r$ for some $q_1, q_2 \in \mathbb{Z}$

and $0 \leq r < m$ by the division algorithm.

It follows $a - b = (q_1 - q_2)m$.

($\Leftarrow$) Suppose $m \mid a - b$.

Again write $a = q_1m + r_1$, $b = q_2m + r_2$

for some $q_1, q_2 \in \mathbb{Z}$, $0 \leq r_1, r_2 < m$.

Then $a - b = (q_1 - q_2)m + r_1 - r_2$

which implies $(a - b) - (q_1 - q_2)m = r_1 - r_2$

So $m \mid r_1 - r_2$. But $-m < r_1 - r_2 < m$

It follows $r_1 = r_2$ and

$a \equiv b \pmod{m}$.

lem 3 : let $m > 0$. Then any integer is congruent to exactly one of $\{ 0, 1, 2, \dots, m-1 \}$.

The modulus operation is compatible with arithmetics on $\mathbb{Z}$.

Prop 4 : let $m > 0$. Suppose $a, b, c, e, f \in \mathbb{Z}$.

Then, (1) if $a \equiv b \pmod{m}$, $b \equiv c \pmod{m}$

then $a \equiv c \pmod{m}$.

(2) if $a \equiv e \pmod{m}$, $b \equiv f \pmod{m}$

then $a+b \equiv e+f \pmod{m}$.

$ab \equiv ef \pmod{m}$.

Remark 5: Be careful that we don't

have a good notion of division (yet)

In particular, we don't have $\frac{1}{2}$ to

multiply 2 yielding 1. But a version

of the cancellation law holds.

Def 6. Let $a, b \in \mathbb{Z}$. We say

a, b are relatively prime if $(a, b) = 1$.

Lemma 7: Suppose $a \mid bc$ and a, b

relatively prime. Then $a \mid c$.

proof. Since $(a, b) = 1$, by Bezout's

theorem. $\exists x, y \in \mathbb{Z}$ s.t. $ax + by = 1$

So $axc + byc = 1$. Note $a \mid bc$

implies $a \mid byc$. Clearly $a \mid axc$.

So $a \mid axc + byc$ which is $a \mid 1$.

Cor 8. Let $r, m > 0$ be relatively prime

Suppose $a, b \in \mathbb{Z}$ such that

$ra \equiv rb \pmod{m}$. Then $a \equiv b \pmod{m}$.

Remark 9. We may rewrite Cor 8

as $r(a - b) \equiv 0 \pmod{m}$.

implies $a - b \equiv 0 \pmod m$ if $(r, m) = 1$.

This should remind us that

for $a, b, n \in \mathbb{Z}$, if $(a - b)n = 0$

and $n \neq 0$, then $a - b = 0$.

An equation of the form $ax + my = b$

for $a, b, m \in \mathbb{Z}$ is called a linear

diophantine equation in two variables

Note that $ax + my = b$ has integer

solution iff $ax \equiv b \pmod m$.

So to solve linear diophantine equation

motivates us to study congruence equation.

Theorem 10: Let $a, b, m \in \mathbb{Z}$, and $m > 0$.

The equation $ax \equiv b \pmod{m}$

has integer solution iff $(a, m) \mid b$.

proof: $(\Rightarrow)$ Suppose x_0 is an integer

solution to $ax \equiv b \pmod{m}$. Then $\exists y \in \mathbb{Z}$

s.t. $ax - b = my$ i.e. $ax - my = b$

As $(a, m) \mid ax$, $(a, m) \mid my$,

we conclude $(a, m) \mid b$.

($\Leftarrow$) Let $d = (a, m)$. and suppose $b = dq$

for some $q \in \mathbb{Z}$. By Bezout's theorem,

write $d = ax_0 + my_0$ for $x_0, y_0 \in \mathbb{Z}$.

Then. $b = aqx_0 + m qy_0$

which implies $a(qx_0) \equiv b \pmod{m}$.

i.e. qx_0 is an integer solution

to the congruence equation $ax \equiv b \pmod{m}$

Def 11. Let $m > 0$. and $a \in \mathbb{Z}$. The

congruence class of a modulus m is

the set of all integers that is

congruent to a modulus m , which is denoted by $[a]$. We also call a an representative element of $[a]$.

Lemma 12: Let $m > 0$, and $(a, m) = 1$

Then $\{0, a, 2a, \dots, (m-1)a\}$ is a complete list of representative of the congruence class modulus m .

That is
$$\bigsqcup_{0 \leq n \leq m-1} [na] = \mathbb{Z}.$$

proof : There are at most m congruence classes because naturally $\{0, 1, 2, \dots, m-1\}$ is a complete list of representatives.

There are m elements in the set $\{0, a, 2a, \dots, (m-1)a\}$.

So it suffices to show for $0 \leq i < j \leq m-1$ $[ia] \neq [ja]$.

Suppose $[ia] = [ja]$.

Then $ia \equiv ja \pmod{m}$.

$$\therefore m \mid (i-j)a \Rightarrow m \mid i-j.$$

But $-m < i-j < m$ so $i=j$.

The good thing of the congruence classes is that it is a language that reduces some complexity by treating the integers having the same remainders as one.

Lemma 13. All solution to $ax \equiv b \pmod{m}$ if exists, lies in the same congruence class mod $\frac{m}{(a, m)}$

Cor 14. $ax \equiv b \pmod{m}$ has integer solution iff $(a, m) \mid b$. If x_0 is a solution then $\left\{ x_0 + k \cdot \frac{m}{(a, m)} : k \in \mathbb{Z} \right\}$ are all the solutions

proof of lemma 13:

Let x, x' be two solutions to $ax \equiv b \pmod{m}$

Then $ax \equiv b \pmod{m}$

$$ax' \equiv b \pmod{m}$$

implies $a(x - x') \equiv 0 \pmod{m}$

i.e. $m \mid a(x - x')$. So $\frac{m}{(a, m)} \mid x - x'$

i.e. $x \equiv x' \pmod{\frac{m}{(a, m)}}$

Clearly if $ax \equiv b \pmod{m}$

$$\begin{aligned} \text{then } a\left(x + k \cdot \frac{m}{(a, m)}\right) &= ax + \frac{a \cdot k \cdot m}{(a, m)} \\ &= ax + \frac{a \cdot k}{(a, m)} m \equiv ax \equiv b \pmod{m}. \end{aligned}$$

because $\frac{a \cdot k}{(a, m)} \in \mathbb{Z}$.